

Challenge (Habanero)

- Q1.** What is the factored form of the quadratic equation $x^2 + 5x + 6 = 0$?
- (A) $(x + 2)(x + 3)$
(B) $(x - 2)(x - 3)$
(C) $(x + 2)(x - 3)$
(D) $(x - 2)(x + 3)$
(E) $(x + 1)(x + 4)$
- Q2.** A hiker is traveling at a rate of $2x^2 - 5x - 3$ miles per hour. If $x = 2$, what is the hiker's rate?
- (A) $2(2)^2 - 5(2) - 3$
(B) $2(2)^2 + 5(2) - 3$
(C) $2(2)^2 - 5(2) + 3$
(D) $2(2)^2 + 5(2) + 3$
(E) $2(2)^2 - 5(2) - 2$
- Q3.** Solve for x : $x^2 + 2x - 15 = 0$
- (A) $x = -5, x = 3$
(B) $x = -3, x = 5$
(C) $x = -2, x = 7$
(D) $x = -7, x = 2$
(E) $x = -1, x = 9$
- Q4.** A car uses $x^2 - 4x + 4$ gallons of fuel per hour. If $x = 3$, how much fuel is used?
- (A) $3^2 - 4(3) + 4$ gallons
(B) $3^2 + 4(3) + 4$ gallons
(C) $3^2 - 4(3) - 4$ gallons
(D) $3^2 + 4(3) - 4$ gallons
(E) $3^2 - 4(3) + 2$ gallons
- Q5.** What is the factored form of the quadratic equation $x^2 - 7x + 12 = 0$?
- (A) $(x - 3)(x - 4)$
(B) $(x + 3)(x + 4)$
(C) $(x - 3)(x + 4)$
(D) $(x + 3)(x - 4)$
(E) $(x - 2)(x - 5)$
- Q6.** The population of a certain animal species is given by the quadratic equation $x^2 + 2x - 6 = 0$. What are the possible values of x ?
- (A) $x = -1 - \sqrt{7}, x = -1 + \sqrt{7}$
(B) $x = -1 - \sqrt{5}, x = -1 + \sqrt{5}$
(C) $x = -3, x = 2$
(D) $x = -2, x = 3$
(E) $x = -1 - \sqrt{3}, x = -1 + \sqrt{3}$
- Q7.** Solve for x : $x^2 - 9x + 20 = 0$
- (A) $x = 4, x = 5$
(B) $x = 5, x = 4$
(C) $x = -4, x = -5$
(D) $x = 9, x = 2$
(E) $x = 1, x = 8$
- Q8.** The area of a rectangular park is given by the quadratic equation $x^2 - 5x - 6 = 0$. What are the possible values of x ?
- (A) $x = -1, x = 6$
(B) $x = -6, x = 1$
(C) $x = 2, x = 3$
(D) $x = -2, x = -3$
(E) $x = -3, x = 2$

Open Ended Questions (FRQ)

- Q9.** A plane uses $x^2 + 2x - 8$ gallons of fuel per minute. If the plane flies for 3 minutes, how much fuel is used in total?
- Q10.** The population of a certain city is given by the quadratic equation $x^2 + 4x + 4 = 0$. Solve for x and explain the meaning of the solutions in the context of the city's population.